

Name: Ayushi Shringi

SID4: 0036

DOMAIN_ID: 4

Domain Entity: Open-source package vulnerabilities

Port: 8036

Prefix: s0036

SEED: 0036

VERIFY_SEED: 260036

Python Version: Python 3.12.14

Model: qwen3:4b

Hardware: MacBook Pro

Chip: Apple M5 Pro

Memory: 24 GB

Operating System: macOS Tahoe 26.5.1

GitHub Repository: <https://github.com/ayushishringi/data260-0036/tree/hw2>

Code Commit Hash: 0d6ab51649e8c24a1af9bc107ca9debf0f8e9a13

Git Tag: hw2

```

<body>
  <main class="card">
    <p class="eyebrow">DATA-260 · HW2 · DOMAIN_ID 4</p>

    <h1>Vulnerability Reports</h1>

    <p class="lede">
      Add, update, delete, and search open-source package vulnerability
      reports.
    </p>

    <section aria-labelledby="search-heading">
      <h2 id="search-heading">Search reports</h2>

      <form id="search-form" class="search-row">
        <label class="sr-only" for="search-input">
          Search by package name or affected version
        </label>

        <input
          id="search-input"
          type="search"
          placeholder="Search package or version"
        />

        <button type="submit">Search</button>
        <button type="button" id="clear-search" class="secondary-button">
          Clear
        </button>
      </form>
    </section>

    <section aria-labelledby="reports-heading">
      <div class="section-heading">
        <h2 id="reports-heading">Current reports</h2>
        <span id="report-count" class="count-badge">0 reports</span>
      </div>

      <p id="loading-state" class="state-message" hidden>
        Loading reports...
      </p>

      <p id="empty-state" class="state-message" hidden>
        No vulnerability reports found.
      </p>

      <p id="error-state" class="state-message error-message" hidden>
        Unable to load reports.
      </p>

      <div id="report-list" class="report-list"></div>
    </section>
  </main>
</body>

```

```
223 .sr-only {
228     margin: -1px;
229     overflow: hidden;
230     clip: rect(0, 0, 0, 0);
231     white-space: nowrap;
232     border: 0;
233 }
234
235 @media (max-width: 600px) {
236     body {
237         padding: 1rem 0.6rem;
238     }
239
240     .card {
241         padding: 1.2rem;
242         border-radius: 12px;
243     }
244
245     h1 {
246         font-size: 1.65rem;
247     }
248
249     .search-row {
250         grid-template-columns: 1fr;
251     }
252
253     .search-row button {
254         width: 100%;
255     }
256
257     .section-heading {
258         align-items: flex-start;
259         flex-direction: column;
260         gap: 0.5rem;
261     }
262
263     .management-actions {
264         flex-direction: column;
265     }
266
267     .management-actions button {
268         width: 100%;
269     }
270 }
```

> @med

```
16
17 function showState({ loading = false, empty = false, error = false }) {
18   loadingState.hidden = (parameter) empty: boolean
19   emptyState.hidden = !empty;
20   errorState.hidden = !error;
21 }
22
23 function showOutput(message) {
24   output.hidden = false;
25   output.textContent = message;
26 }
27
28 function formatDate(dateString) {
29   return new Date(dateString).toLocaleString();
30 }
31
32 function renderReports(reports) {
33   reportList.replaceChildren();
34   reportCount.textContent = `${reports.length} ${
35     reports.length === 1 ? "report" : "reports"
36   }`;
37
38   if (reports.length === 0) {
39     showState({ empty: true });
40     return;
41   }
42
43   showState({});
```

Vulnerability Reports

Add, update, delete, and search open-source package vulnerability reports.

Search reports

SearchClear

Current reports

2 reports

curl — ID 1

Affected version: < 8.8.0 | Severity: high

A vulnerability in curl may allow an attacker to bypass security checks through a specially crafted network request.

Submitted by: security@curl.se

Submitted: 9/1/2026, 5:00:00 AM

Loadlash — ID 2

Affected version: 3.2.3 | Severity: high

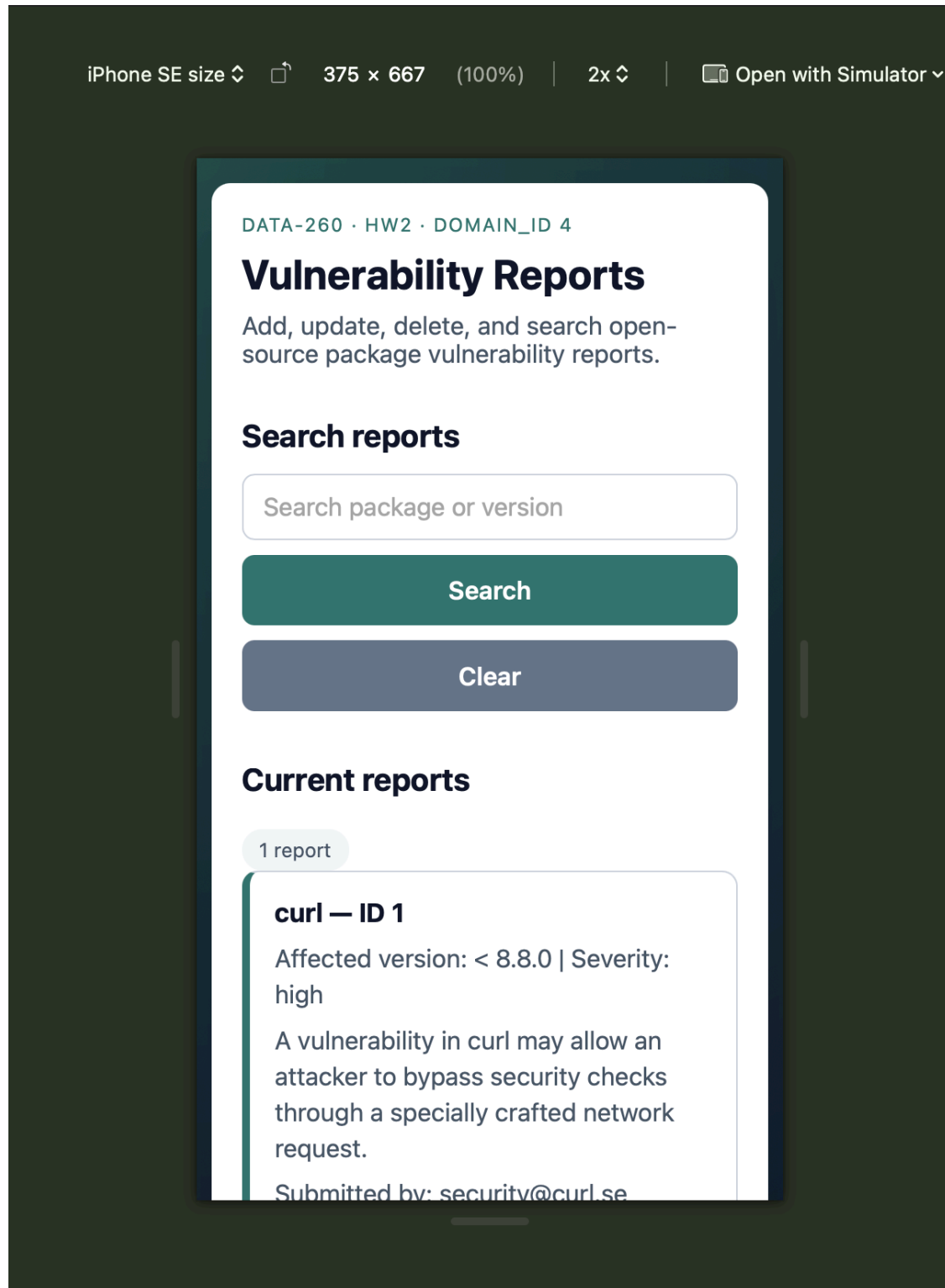
This is a security issue which needs attention right now because the server is down and all.

Submitted by: teste@gmail.com

Submitted: 9/13/2026, 7:50:33 PM

Update record ID 1Delete highest ID

Responsive mobile layout at 375px



iPhone SE size 375 x 667 (100%) | 2x | Open with Simulator

Delete highest ID

Add a vulnerability report

Package name

Example: lodash

Affected version

Example: < 4.17.21

Submitter email

you@example.com

Vulnerability description

Describe the issue in more than 25 characters.

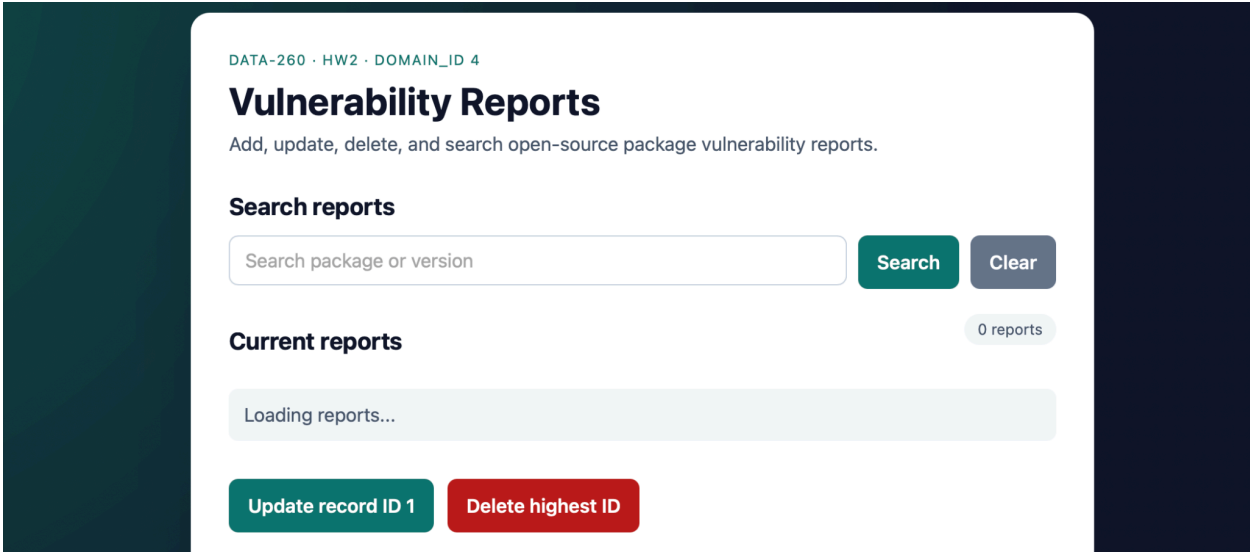
Severity

Select a severity

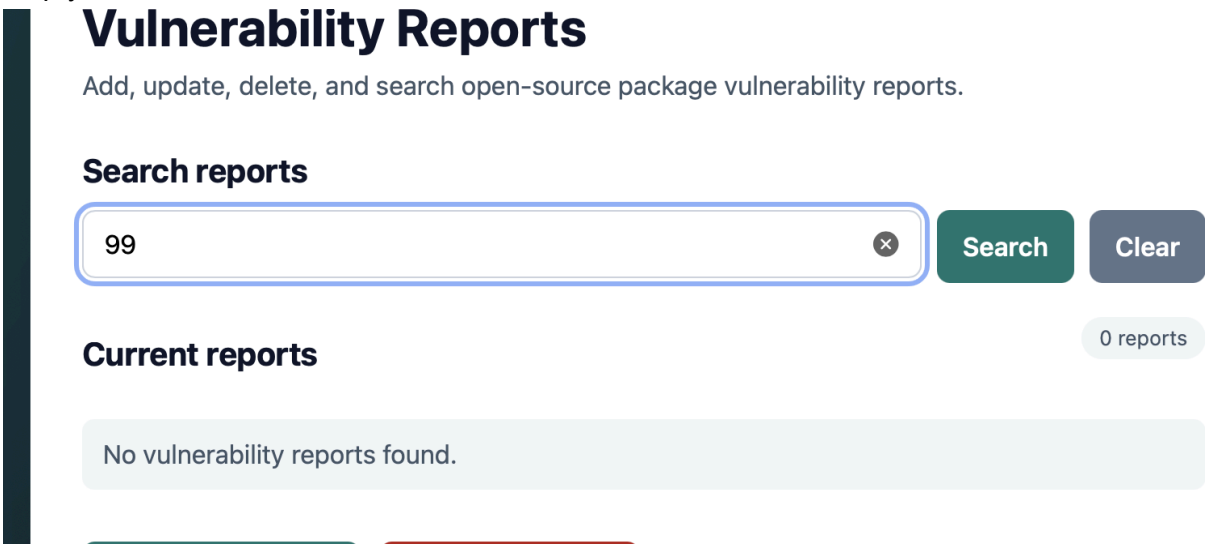
☐ I agree to the terms and conditions.

Add vulnerability report

Loading state



Empty Search Result



Backend Error

DATA-260 · HW2 · DOMAIN_ID 4

Vulnerability Reports

Add, update, delete, and search open-source package vulnerability reports.

Search reports

SearchClear

Current reports

Unavailable

Unable to load reports.

Adding New Report

```

@app.post("/api/reports", response_model=VulnerabilityReport, status_code=201)
def create_report(report: VulnerabilityReportInput) -> dict:
    records = load_records()
    next_id = max((record["id"] for record in records), default=0) + 1

    new_record = {
        "id": next_id,
        **report.model_dump(),
        "submissionDate": now_iso(),
    }

    records.append(new_record)
    save_records(records)

    return new_record

@app.put("/api/reports/1", response_model=VulnerabilityReport)
def update_first_report(report: VulnerabilityReportInput) -> dict:
    records = load_records()

    for index, record in enumerate(records):
        if record["id"] == 1:
            updated_record = {
                "id": 1,
                **report.model_dump(),
                "submissionDate": record["submissionDate"],
            }

            records[index] = updated_record
            save_records(records)

    return updated_record

```

☐ I agree to the terms and conditions.

Add vulnerability report

Added vulnerability report with ID 3.

Current reports

3 reports

curl — ID 1

Affected version: < 8.8.0 | Severity: high

A vulnerability in curl may allow an attacker to bypass security checks through a specially crafted network request.

Submitted by: security@curl.se

Submitted: 9/1/2026, 5:00:00 AM

Loadlash — ID 2

Affected version: 3.2.3 | Severity: high

This is wiser issuer which needs attention right now because server is down and alll.

Submitted by: teste@gmail.com

Submitted: 9/13/2026, 7:50:33 PM

django — ID 3

Affected version: 3.1.45 | Severity: critical

A crafted request may bypass security checks and expose sensitive application data.

Submitted by: test@gmail.com

Submitted: 9/13/2026, 8:08:30 PM

Update record ID 1

Delete highest ID

Updating Record

```
@app.put("/api/reports/1", response_model=VulnerabilityReport)
def update_first_report(report: VulnerabilityReportInput) -> dict:
    records = load_records()

    for index, record in enumerate(records):
        if record["id"] == 1:
            updated_record = {
                "id": 1,
                **report.model_dump(),
                "submissionDate": record["submissionDate"],
            }

            records[index] = updated_record
            save_records(records)

            return updated_record

    raise HTTPException(
        status_code=404,
        detail="Record with ID 1 was not found.",
    )
```

SearchClear

Current reports

3 reports

curl — ID 1

Affected version: < 8.8.0 | Severity: high

A vulnerability in curl may allow an attacker to bypass security checks through a specially crafted network request.

Submitted by: security@curl.se

Submitted: 9/1/2026, 5:00:00 AM

Loadlash — ID 2

Affected version: 3.2.3 | Severity: high

This is wiser issuer which needs attention right now because server is down and alll.

Submitted by: teste@gmail.com

Submitted: 9/13/2026, 7:50:33 PM

django — ID 3

Affected version: 3.1.45 | Severity: critical

A crafted request may bypass security checks and expose sensitive application data.

Submitted by: test@gmail.com

Submitted: 9/13/2026, 8:08:30 PM

Update record ID 1Delete highest ID

Record ID 1 was updated successfully.

Deleting Record

```
@app.delete("/api/reports/highest", response_model=VulnerabilityReport)
def delete_highest_report() -> dict:
    records = load_records()

    if not records:
        raise HTTPException(
            status_code=404,
            detail="No records exist.",
        )

    highest_record = max(records, key=lambda record: record["id"])
    records.remove(highest_record)
    save_records(records)

    return highest_record
```

Current reports

2 reports

curl — ID 1

Affected version: < 8.8.0 | Severity: high

A vulnerability in curl may allow an attacker to bypass security checks through a specially crafted network request.

Submitted by: security@curl.se

Submitted: 9/1/2026, 5:00:00 AM

Loadlash — ID 2

Affected version: 3.2.3 | Severity: high

This is wiser issuer which needs attention right now because server is down and all.

Submitted by: teste@gmail.com

Submitted: 9/13/2026, 7:50:33 PM

Update record ID 1

Delete highest ID

Deleted record ID 3.

Search Report

```

@app.get("/api/reports", response_model=list[VulnerabilityReport])
def list_reports(
    search: str | None = Query(default=None),
) -> list[dict]:
    records = load_records()

    if not search or not search.strip():
        return records

    search_text = search.strip().lower()

    return [
        record
        for record in records
        if search_text in record["packageName"].lower()
        or search_text in record["affectedVersion"].lower()
    ]

```

```

searchForm.addEventListener("submit", async (event) => {
    event.preventDefault();
    await loadReports(searchInput.value);
});

```

Search reports

×

Search

Clear

Current reports

1 report

Loadlash — ID 2

Affected version: 3.2.3 | Severity: high

This is a wider issue which needs attention right now because the server is down and all.

Submitted by: teste@gmail.com

Submitted: 9/13/2026, 7:50:33 PM

Shared LangGraph Agent State

```
class AgentState(TypedDict, total=False):  
    title: str  
    content: str  
    llm: Any  
    turn_ceiling: int  
    turn_count: int  
    planner_attempts: int  
    planner_proposal: dict[str, Any]  
    reviewer_feedback: dict[str, Any]  
    validation_error: str  
    status: str
```

Planner

```

01
02 def planner_node(state: AgentState) -> dict[str, Any]:
03     print("--- NODE: Planner ---")
04
05     planner_attempts = state.get("planner_attempts", 0) + 1
06
07     previous_error = state.get("validation_error", "")
08     previous_feedback = state.get("reviewer_feedback", {})
09
10     correction = ""
11
12     if previous_error:
13         correction += (
14             "\nPrevious validation error. Fix it carefully:\n"
15             + previous_error
16         )
17
18     if previous_feedback:
19         correction += (
20             "\nReviewer feedback from the previous attempt:\n"
21             + json.dumps(previous_feedback)
22         )
23
24     source = (
25         f"Title: {state['title']}\n\n"
26         f"Body:\n{state['content']}\n"
27         f"{correction}"
28     )
29
30     result = state["llm"].complete(
31         [
32             {"role": "system", "content": PLANNER_SYSTEM},
33             {"role": "user", "content": source},
34         ],
35         temperature=0.2,
36         format=PlannerOutput.model_json_schema(),
37     )
38
39     try:
40         proposal = PlannerOutput.model_validate(extract_json(result.text))
41
42         return {
43             "planner_proposal": proposal.model_dump(),

```

Reviewer and /supervisor

```

def reviewer_node(state: AgentState) -> dict[str, Any]:
    print("---- NODE: Reviewer ----")

    source = (
        f"Title: {state['title']}\n\n"
        f"Body:\n{state['content']}\n\n"
        f"Planner proposal:\n"
        f"{json.dumps(state['planner_proposal'])}"
    )

    result = state["llm"].complete(
        [
            {"role": "system", "content": REVIEWER_SYSTEM},
            {"role": "user", "content": source},
        ],
        temperature=0.0,
        format=ReviewOutput.model_json_schema(),
    )

    try:
        review = ReviewOutput.model_validate(extract_json(result.text))
        feedback = review.model_dump()

    except (ValidationError, ValueError, TypeError) as exc:
        feedback = {
            "approved": False,
            "feedback": f"Reviewer output was invalid: {exc}",
        }

    return {"reviewer_feedback": feedback}

```

```
def supervisor_node(state: AgentState) -> dict[str, Any]:  
    print("--- NODE: Supervisor ---")  
  
    next_turn = state.get("turn_count", 0) + 1  
    ceiling = state.get("turn_ceiling", 10)  
    feedback = state.get("reviewer_feedback", {})  
  
    status = "running"  
  
    if feedback.get("approved") is True:  
        status = "completed"  
    elif next_turn > ceiling:  
        status = "abandoned"  
  
    return {  
        "turn_count": next_turn,  
        "status": status,  
    }
```

Supervisor Routing and Graph Assembly

```

0  def router_logic(state: AgentState) -> str:
1      ceiling = state.get("turn_ceiling", 10)
2
3      if state.get("status") in {"completed", "abandoned"}:
4          return "end"
5
6      if state.get("validation_error"):
7          if state.get("turn_count", 0) <= ceiling:
8              return "planner"
9
10         return "end"
11
12     feedback = state.get("reviewer_feedback", {})
13
14     if feedback.get("approved") is True:
15         return "end"
16
17     if feedback and state.get("turn_count", 0) < ceiling:
18         return "planner"
19
20     if not state.get("planner_proposal"):
21         return "planner"
22
23     return "reviewer"
24
25
26 def after_planner_route(state: AgentState) -> str:
27     if state.get("validation_error"):
28         return "supervisor"
29
30     return "reviewer"
31
32
33 def build_graph():
34     workflow = StateGraph(AgentState)
35
36     workflow.add_node("supervisor", supervisor_node)
37     workflow.add_node("planner", planner_node)
38     workflow.add_node("reviewer", reviewer_node)
39
40     workflow.add_edge(START, "supervisor")
41
42     workflow.add_conditional_edges(
43         "supervisor",

```

```

        router_logic,
        {
            "planner": "planner",
            "reviewer": "reviewer",
            "end": END,
        },
    )

    workflow.add_conditional_edges(
        "planner",
        after_planner_route,
        {
            "supervisor": "supervisor",
            "reviewer": "reviewer",
        },
    )

    workflow.add_edge("reviewer", "supervisor")

    return workflow.compile()

```

```

def main() -> int:
    parser = argparse.ArgumentParser()
    parser.add_argument("--title", required=True)
    parser.add_argument("--content", required=True)
    parser.add_argument("--model", default=None)
    parser.add_argument("--turn-ceiling", type=int, default=10)
    args = parser.parse_args()

    client = ModelClient(model=args.model) if args.model else ModelClient()

    initial_state: AgentState = {
        "title": args.title,
        "content": args.content,
        "llm": client,
        "turn_ceiling": args.turn_ceiling,
        "turn_count": 0,
        "planner_attempts": 0,
        "planner_proposal": {},
        "reviewer_feedback": {},
        "validation_error": "",
        "status": "running",
    }

    graph = build_graph()
    final_state = dict(initial_state)

    for update in graph.stream(initial_state):
        print(json.dumps(update, indent=2, default=str))

```

```

((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % .venv/bin/python3 agents_graph.py \
  --title "OpenSSL certificate validation vulnerability" \
  --content "A vulnerable OpenSSL version may allow an attacker to bypass certificate validation and
intercept encrypted network traffic." \
  --turn-ceiling 10
└─
--- NODE: Reviewer ---
{
  "reviewer": {
    "reviewer_feedback": {
      "approved": true,
      "feedback": "Tags are relevant and summary is factual and concise."
    }
  }
}
--- NODE: Supervisor ---
{
  "supervisor": {
    "turn_count": 2,
    "status": "completed"
  }
}

Graph execution finished.
{
  "title": "OpenSSL certificate validation vulnerability",
  "content": "A vulnerable OpenSSL version may allow an attacker to bypass certificate validation and
intercept encrypted network traffic.",
  "llm": "<model_client.ModelClient object at 0x10a697200>",
  "turn_ceiling": 10,
  "turn_count": 2,
  "planner_attempts": 1,
  "planner_proposal": {
    "tags": [
      "openssl vulnerability",
      "certificate validation",
      "network traffic interception"
    ],
    "summary": "OpenSSL vulnerability enables bypassing certificate validation to intercept encrypted
traffic."
  },
  "reviewer_feedback": {
    "approved": true,
    "feedback": "Tags are relevant and summary is factual and concise."
  },
  "validation_error": "",
  "status": "completed"
}

```

Planner Output Validation

```

class PlannerOutput(BaseModel):
    tags: list[str]
    summary: str

    @field_validator("tags")
    @classmethod
    def validate_tags(cls, value: list[str]) -> list[str]:
        if len(value) != 3:
            raise ValueError("There must be exactly three tags.")

        for tag in value:
            if not isinstance(tag, str):
                raise ValueError("Every tag must be a string.")

            if not 3 <= len(tag.strip()) <= 30:
                raise ValueError(
                    "Every tag must contain between 3 and 30 characters."
                )

        return [tag.strip() for tag in value]

    @field_validator("summary")
    @classmethod
    def validate_summary(cls, value: str) -> str:
        word_count = len(re.findall(r"\b[\w'-]+\b", value))

        if word_count > 25:
            raise ValueError("The summary must contain at most 25 words.")

        return value.strip()

```

Thirty Run Schema Validation

```

def run_schema_experiment(client: ModelClient) -> None:
    print("Running 30 schema-validation experiments...")

    case = load_case("schema_input.json")
    graph = build_graph()
    rows = []

    for run_number in range(1, 31):
        row = run_once(graph, client, case, 10, run_number)
        rows.append(row)

        print(
            f"Run {run_number}/30: "
            f"{row['classification']} | "
            f"attempts={row['planner_attempts']} | "
            f"latency={row['latency_ms']} ms",
            flush=True,
        )

    save_results("schema_30_runs", rows)

```

```

def run_ceiling_experiment(client: ModelClient) -> None:
    print("Running 20 experiments with ceiling 2...")
    case = load_case("schema_input.json")
    graph = build_graph()

    ceiling_2_rows = []

    for run_number in range(1, 21):
        row = run_once(graph, client, case, 2, run_number)
        ceiling_2_rows.append(row)

        print(
            f"Ceiling 2, run {run_number}/20: "
            f"{row['classification']} | "
            f"latency={row['latency_ms']} ms",
            flush=True,
        )

    save_results("ceiling_2_20_runs", ceiling_2_rows)

    print("Running 20 experiments with ceiling 10...")

```

```
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % cat reports/hw02/raw/schema_30_runs_metrics.json
{
  "runs": 30,
  "completed": 30,
  "completion_rate": 1.0,
  "mean_latency_ms": 848.42,
  "classification_counts": {
    "Valid first attempt": 30
  }
}
```

Turn Ceiling comparison

```

def run_ceiling_experiment(client: ModelClient) -> None:
    print("Running 20 experiments with ceiling 2...")
    case = load_case("schema_input.json")
    graph = build_graph()

    ceiling_2_rows = []

    for run_number in range(1, 21):
        row = run_once(graph, client, case, 2, run_number)
        ceiling_2_rows.append(row)

        print(
            f"Ceiling 2, run {run_number}/20: "
            f"{row['classification']} | "
            f"latency={row['latency_ms']} ms",
            flush=True,
        )

    save_results("ceiling_2_20_runs", ceiling_2_rows)

    print("Running 20 experiments with ceiling 10...")

    ceiling_10_rows = []

    for run_number in range(1, 21):
        row = run_once(graph, client, case, 10, run_number)
        ceiling_10_rows.append(row)

        print(
            f"Ceiling 10, run {run_number}/20: "
            f"{row['classification']} | "
            f"latency={row['latency_ms']} ms",
            flush=True,
        )

    save_results("ceiling_10_20_runs", ceiling_10_rows)

```

```

• ((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % cat reports/hw02/raw/ceiling_2_20_runs_metrics.json
cat reports/hw02/raw/ceiling_10_20_runs_metrics.json
{
  "runs": 20,
  "completed": 20,
  "completion_rate": 1.0,
  "mean_latency_ms": 855.11,
  "classification_counts": {
    "Valid first attempt": 20
  }
}
{
  "runs": 20,
  "completed": 20,
  "completion_rate": 1.0,
  "mean_latency_ms": 840.64,
  "classification_counts": {
    "Valid first attempt": 20
  }
}
}

```

```

• ((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % cat reports/hw02/raw/adversarial_5_runs_metrics.json
{
  "runs": 5,
  "completed": 0,
  "completion_rate": 0.0,
  "mean_latency_ms": 2069.5,
  "classification_counts": {
    "Hit turn ceiling": 5
  }
}
}

```

```

• ((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % make verify-hw02
.venv/bin/python3 scripts/verify_hw02.py
--- NODE: Supervisor ---
--- NODE: Planner ---
--- NODE: Reviewer ---
--- NODE: Supervisor ---
{
  "passed": true,
  "homework": "DATA-260 HW2",
  "SID4": "0036",
  "PORT_BASE": 8036,
  "PREFIX": "s0036",
  "SEED": 36,
  "VERIFY_SEED": 260036,
  "DOMAIN_ID": 4,
  "model": "qwen3:4b",
  "commit_hash": "7774a82f4f1e93067e998d0980d49098717294c3",
  "checks": {
    "required_files": {
      "index.html": true,
      "app.js": true,
      "styles.css": true,
      "requirements.txt": true,
      "backend/main.py": true,
      "agents_graph.py": true,
      "scripts/run_hw02_experiments.py": true,
      "scripts/verify_hw02.py": true,
      "reports/hw02/RUN_LOG.txt": true,
      "reports/hw02/METRICS.md": true,

```